

1. Duality in Linear Programs

(a) ?

(b) (i) $x_i = 1$ means $v_i \in S$

$x_i = 0$ means $v_i \in T$

if x_i, x_j variables weren't introduced
optimal $b=0$, we can't find a cut.

$$(ii) f_{LP}^* \leq f_{ILP}^*$$

(iii)

$$\sum_{(i,j) \in E} f_{ij} (b_{ij} - x_i + x_j) + \sum_{(i,j) \in E} y_{ij} b_{ij} + W (x_s - x_t - 1)$$

$$= \sum_{(i,j) \in E} b_{ij} (f_{ij} + y_{ij}) + \sum_{(i,j) \in E} f_{ij} (-x_i + x_j) + W (x_s - x_t) \geq W$$

$$f_{ij} + y_{ij} = c_{ij} \quad , \quad f_{ij} \geq 0 \quad , \quad y_{ij} \geq 0$$

$$\sum_{(i,j) \in E} f_{ij} (-x_i + x_j) + W (x_s - x_t) = 0 \quad (\text{to cancel out } x_i)$$

$$= \sum_{k \notin \{s, t\}} \sum_{ik} f_{ik} (-x_i + x_k) + \sum_{kj} f_{kj} (-x_k + x_j)$$

$$+ \sum_{sj} f_{sj} (-x_s + x_j) + \sum_{it} f_{it} (-x_i + x_t)$$

$$+ W (x_s - x_t)$$

$$\begin{aligned}
&= \sum_{k \notin \{s, t\}} x_k \left(\sum_{i:k} f_{ik} - \sum_{j:k} f_{kj} \right) + \left(\sum_{k \neq t} \sum_{i:k} f_{ik} + \sum_{i:t} f_{it} \right) (-x_i) \\
&+ \left(\sum_{k \neq s} \sum_{j:k} f_{kj} + \sum_{s:j} f_{sj} \right) x_j + \left(\sum_{s:j} f_{sj} - W \right) x_s \\
&+ \left(\sum_{i:t} f_{it} - W \right) x_t \\
&= \sum_{k \notin \{s, t\}} x_k \left(\sum_{i:k} f_{ik} - \sum_{j:k} f_{kj} \right) + \cancel{\sum_{i:j} f_{ij} x_i} - \cancel{\sum_{i:j} f_{ij} x_i} \\
&+ \left(\sum_{s:j} f_{sj} - W \right) x_s + \left(\sum_{i:t} f_{it} - W \right) x_t
\end{aligned}$$

To cancel out x_i , $W = \sum_{s:j} f_{sj} = \sum_{i:t} f_{it}$

$$\sum_{i:k} f_{ik} = \sum_{j:k} f_{kj} \quad \text{for } k \notin \{s, t\}$$

So we have the dual problem:

$$\max_{f \in \mathbb{R}^{|E|}} \sum_{(s,j) \in E} f_{sj}$$

$$\text{s.t. } 0 \leq f_{ij} \leq c_{ij} \quad \dots \textcircled{1}$$

$$\sum_{i:k} f_{ik} = \sum_{j:k} f_{kj} \quad \text{for } k \notin \{s, t\} \quad \dots \textcircled{2}$$

(iv) Constrain ① : Number of students on path $V_i \rightarrow V_j$ is non-negative and less than the capacity c_{ij} .

Constrain ② : At each landmark $k \notin \{s, t\}$, no student vanishes

$$(v) \quad f_{\text{dual}}^* \leq f_{LP}^*$$

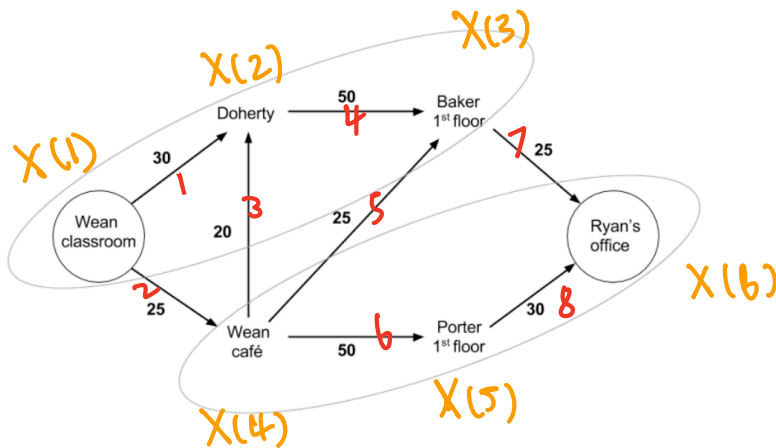
$$(vi) \quad \because f_{\text{dual}}^* = f_{ILP}^*$$

$$\therefore f_{\text{dual}}^* = f_{LP}^* = f_{ILP}^*$$

convex relaxation of ILP is tight actually.

(vii) Barricade: $\{1, 3, 4, 5, 6, 8\}$

max flow = 50



```
clear,clc
```

```
c = [30 25 20 50 25 50 25 30]';
```

```
E = length(c);
```

```
cvx_begin
```

```
variable b(E)
```

```
variable x(6)
```

```
minimize(c'*b)
```

```
subject to
```

```
x(1) - x(6) >= 1;
```

```
b >= 0;
```

```
b(1) >= x(1) - x(2);
```

```
b(2) >= x(1) - x(4);
```

```
b(3) >= x(4) - x(2);
```

```
b(4) >= x(2) - x(3);
```

```
b(5) >= x(4) - x(3);
```

```
b(6) >= x(4) - x(5);
```

```
b(7) >= x(3) - x(6);
```

```
b(8) >= x(5) - x(6);
```

```
cvx_end
```

2. Practice with KKT conditions and duality

(a)

$$(i) \quad u^T x + v^T (Ax - b) \geq 0, \quad u \geq 0$$

$$(u + A^T v)^T x \geq v^T b$$

$$\text{So dual LP: } \max_v v^T b \quad \text{s.t. } A^T v \leq c$$

(ii)

$$\text{Stationary: } c + A^T v - u = 0$$

$$\text{Complementary slackness: } u^T x = 0$$

$$\text{Primal feasibility: } Ax = b, \quad x \geq 0$$

$$\text{dual feasibility: } u \geq 0$$

(iii)

$$\text{Stationary: } c + A^T v - u = \frac{\tau}{x}$$

$$\text{Complementary slackness: } u^T x = 0$$

$$\text{Primal feasibility: } Ax = b, \quad x > 0$$

$$\text{dual feasibility: } u \geq 0$$

(iv) ?

(b)

$$\min_{\beta} \|y - X\beta\|_2^2$$

↓

$$\min_{\beta, z} \|y - z\|_2^2$$

$$\text{s.t. } z = X\beta$$

$$L(\beta, z, v) = \|y - z\|_2^2 + v^T (z - X\beta)$$

$$\frac{\partial L}{\partial \beta} = -X^T v = 0$$

$$\frac{\partial L}{\partial z} = 2(z - y) + v = 0, \quad z = y - \frac{v}{2}$$

$$g(v) = \min_{\beta, z} L(\beta, z, v) = v^T \left(y - \frac{1}{4} v \right)$$

This is not equal to $\|y - v\|_2^2$. ?

$$\text{KKT: } \partial \left(\|y - X\beta\|_2^2 + v^T (z - X\beta) \right) = 0$$

⇓

$$z^* - X\beta^* = 0$$

$$v^* = 0$$

$$X^T (y - X\beta^*) = X^T v^*$$

↓

v^* is the error in the projected space.

3. Convex conjugate and Moreau decomposition

$$\begin{aligned}(a) \quad (th)^*(x) &= \sup_{y \in \text{dom}(th)} x^T y - (th)(y) \\&= \sup_{y \in \text{dom}(h)} x^T y - th(y) \\&= t \sup_{y \in \text{dom}(h)} \left(\frac{x}{t}\right)^T y - h(y) \\&= th^*\left(\frac{x}{t}\right)\end{aligned}$$

$$(b) \quad y \in \partial f(x), \quad f(z) \geq f(x) + y^T(z - x) \quad \forall z$$

$$y^T z - f(z) \leq y^T x - f(x)$$

$$f^*(y) \leq y^T x - f(x) \quad (\text{let } z = \arg \sup_z y^T z - f(z))$$

$$f^*(z) \geq z^T x - f(x) \quad \forall z \quad (\text{definition})$$

$$f^*(z) - f^*(y) \geq z^T x - f(x) + f(x) - y^T x = x^T(z - y)$$

$$\text{So } x \in \partial f^*(y)$$

$$y \in \partial f(x) \Rightarrow x \in \partial f^*(y) \Rightarrow y \in \partial f^{**}(x)$$

$$\text{because } f^{**} = f$$

$$\text{So } y \in \partial f(x) \iff x \in \partial f^*(y)$$

$$(c) \text{prox}_{th}(x) = u, \quad x - u \in t \partial h(u)$$

$$\text{prox}_{\frac{1}{t}h^*}\left(\frac{x}{t}\right) = \frac{v}{t}, \quad x - v \in \partial h^*\left(\frac{v}{t}\right)$$

$$x - u \in t \partial h(u)$$

$$u \in \partial h^*\left(\frac{x - u}{t}\right)$$

$$x - (x - u) \in \partial h^*\left(\frac{x - u}{t}\right)$$

$$v = x - u$$

$$\text{prox}_{th}(x) + t \text{prox}_{\frac{1}{t}h^*}\left(\frac{x}{t}\right) = u + v = x$$

$$(d) \text{ dual norm } \|x\|_* = \sup_{\|z\| \leq 1} z^T x$$

$$h^*(x) = \sup_y x^T y - \|y\|$$

if $\|x\|_* \leq 1$, $x^T y \leq \|y\|$, because $\|y\| = \sup_{\|x\|_* \leq 1} x^T y$

equality holds if $y = 0$, so $h^*(x) = 0$

if $\|x\|_* > 1$, $\exists \|y\| \leq 1$ s.t. $x^T y > 1$

$$h^*(x) \geq x^T (ty) - \|ty\| = t(x^T y - \|y\|)$$

when $t \rightarrow \infty$, $h^*(x) \rightarrow \infty$

(e)

$$h(z) = \|z\|_\infty \xleftrightarrow{\text{dual norm}} \|z\|_1$$

$$h^*(z) = \begin{cases} 0, & \|z\|_1 \leq 1 \\ \infty, & \|z\|_1 > 1 \end{cases} = \mathbb{1}_{\{\|z\|_1 \leq 1\}}(z)$$

$$\text{prox}_{th}(x) = x - t \text{prox}_{\frac{1}{t}h^*}\left(\frac{x}{t}\right)$$

$$\begin{aligned} \text{prox}_{\frac{1}{t}h^*}\left(\frac{x}{t}\right) &= \arg\min_z \frac{1}{2} \|z - \frac{x}{t}\|_2^2 + \frac{1}{t} h^*(z) \\ &= \text{Proj}_{\{\|z\|_1 \leq 1\}}\left(\frac{x}{t}\right) \end{aligned}$$

$$\text{So } \text{prox}_{t\|\cdot\|_\infty}(x) = x - t \underbrace{\text{Proj}_{\{\|z\|_1 \leq 1\}}\left(\frac{x}{t}\right)}_{\text{can be computed by a known algorithm.}}$$

4. Support vector machines and duality

Part (a)

(i) yes, because the inequality constraints are affine

$$(ii) \mathcal{L}(\beta, \beta_0, \gamma, \alpha, \mu) = \frac{1}{2} \|\beta\|_2^2 + C_1 \sum_{i \in S_1} \gamma_i + C_2 \sum_{i \in S_2} \gamma_i \\ - \sum_i \mu_i \gamma_i + \sum_i \alpha_i (1 - \gamma_i - y_i (x_i^T \beta + \beta_0))$$

Stationarity:

$$\beta - \sum_i \alpha_i y_i x_i = 0 \quad \text{for } \beta$$

$$\sum_i \alpha_i y_i = 0 \quad \text{for } \beta_0$$

$$C_1 - \mu_i - \alpha_i = 0 \quad i \in S_1 \quad \text{for } \gamma$$

$$C_2 - \mu_i - \alpha_i = 0 \quad i \in S_2$$

Complementary slackness:

$$\mu_i \gamma_i = 0$$

$$\alpha_i (1 - \gamma_i - y_i (x_i^T \beta + \beta_0)) = 0$$

primal and dual feasibility

$$\mu, \alpha \geq 0 \quad \gamma \geq 0 \quad y_i (x_i^T \beta + \beta_0) \geq 1 - \gamma_i$$

$$(iii) \quad g(\alpha, \mu) = \min_{\beta, \beta_0, \gamma} \mathcal{L}(\beta, \beta_0, \gamma, \alpha, \mu)$$

$$= \begin{cases} -\frac{1}{2} \alpha^T \tilde{X} \tilde{X} \alpha + 1^T \alpha & \text{if } y^T \alpha = 0, \quad \alpha_{s_1} = C_{11} - \mu_{s_1}, \quad \alpha_{s_2} = C_{12} - \mu_{s_2} \\ -\infty & \text{otherwise} \end{cases}$$

$$\max_{\alpha} g(\alpha, \mu) \Leftrightarrow \max_{\alpha} -\frac{1}{2} \alpha^T \tilde{X} \tilde{X} \alpha + 1^T \alpha$$

$$\text{s.t. } y^T \alpha = 0$$

$$\tilde{X} = \text{diag}(y) X$$

$$0 \leq \alpha_{s_1} \leq C_{11}$$

$$0 \leq \alpha_{s_2} \leq C_{21}$$

$$(iv) \quad \beta_{\star} = \sum_i \alpha_{\star i} y_i X_i$$

We want to max $1^T \alpha$ while α_{s_i} is

bounded by C_i , $i=1,2$, so $\alpha_{\star}$

is cost-sensitive and so is $\beta_{\star}$.

(V) QP

Part (b) using Matlab quadprog function

$$(i) Z = (\beta, \beta_0, \gamma) \in \mathbb{R}^{p+1+n}$$

$$H = \text{diag}(\underbrace{1, \dots, 1}_p, \underbrace{0, \dots, 0}_{1+n})$$

$$f = (\underbrace{0, \dots, 0}_{p+1}, c_1 \dots c_2 \dots c_2 \dots c_1)$$

$$y_i x_i^T \beta + y_i \beta_0 + \gamma_i \geq 1$$

$$(Ax)_i = \sum_j A_{ij} x_j$$

$$Az \geq 1$$

$$j \in [1, p], \quad A_{i, [1, p]} = y_i x_i^T$$

$$j = p+1, \quad A_{i, p+1} = y_i$$

$$j = [p+2, n], \quad A_{i, [p+2, \text{end}]} = \underbrace{[0, \dots, 1, \dots, 0]}_n$$

$\downarrow$
 i

$$lb = [-\text{inf}, \dots, -\text{inf}, \underbrace{1, \dots, 1}_n]$$

$$C_1 = C_2 = 1$$

$$C_1 = 1, C_2 = 10$$

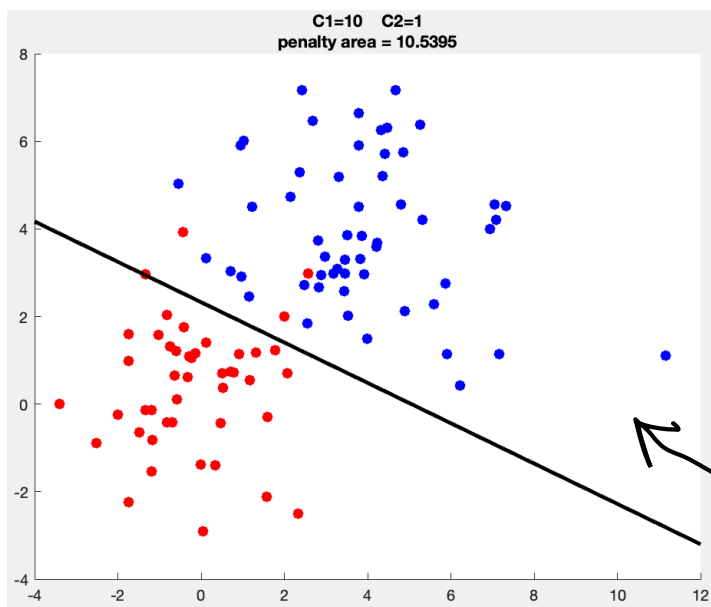
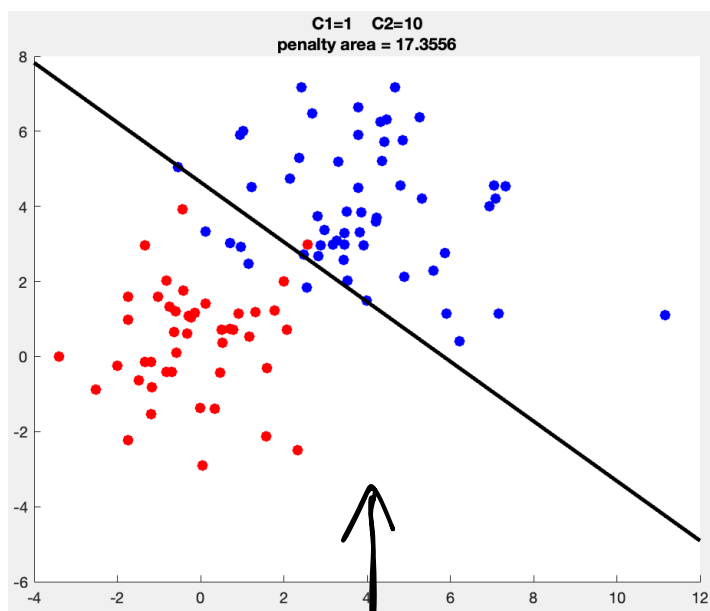
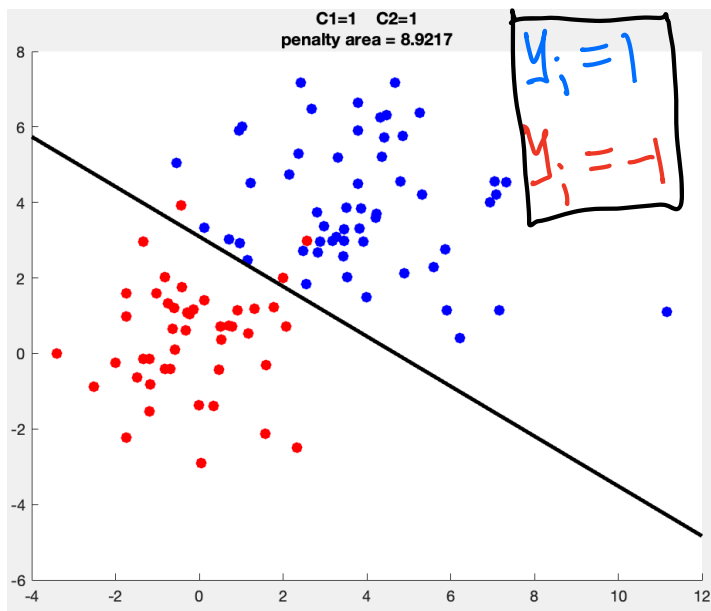
$$C_1 = 10, C_2 = 1$$

$$fval = 9.6044$$

$$fval = 29.8981$$

$$fval = 11.8911$$

(ii)



more penalty on y_i

more penalty on y_i

(iii)

add a $(n+1)$ -th point on the right side

$$y_i (x_i^T \beta + \beta_0) > 1, \quad \gamma_i = 0, \quad i = n+1$$

$$\text{KKT said } \alpha_i (1 - \gamma_i - y_i (x_i^T \beta + \beta_0)) = 0$$

$$\text{so } \alpha_i = 0$$

$$\beta = \sum_i \alpha_i y_i x_i \text{ won't change.}$$

(iv) dual problem using quadprog:

$$H = \tilde{X} \tilde{X}^T, \quad \tilde{X} = \text{diag}(y) X$$

$$f = 1$$

$$A = []$$

$$b = []$$

$$A_{eq} = y^T$$

$$b_{eq} = 0$$

$$lb = 0$$

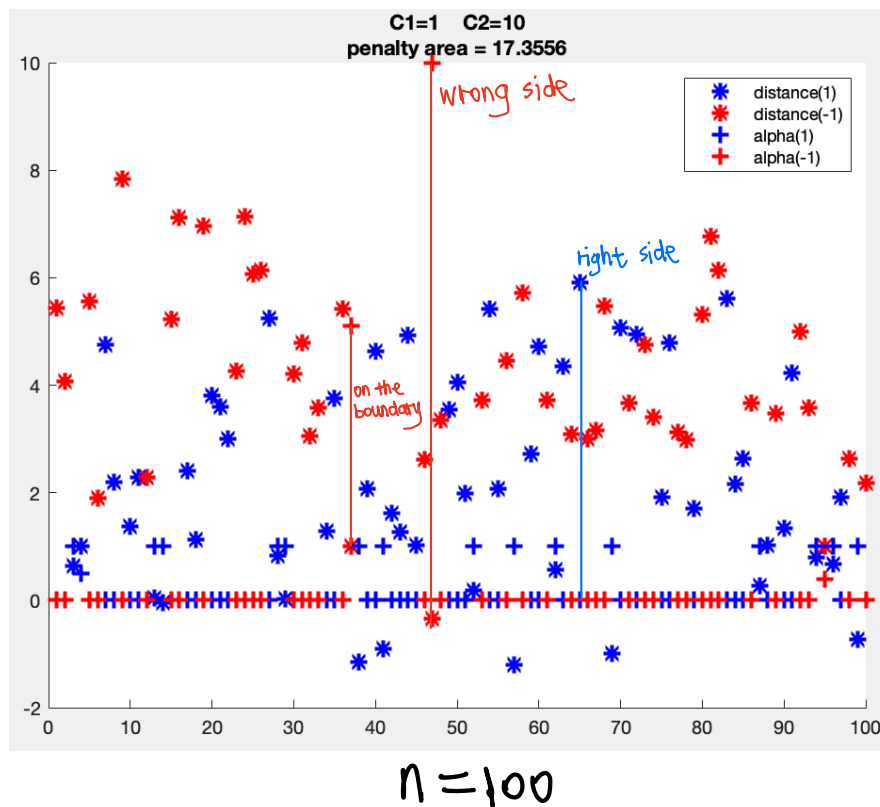
$$ub = [\dots c_1, \dots c_2, \dots, c_1, \dots c_2, \dots]$$

primal fval = dual fval

$\alpha_i = 0 \rightarrow$ right side

$\alpha_i = C_k \rightarrow$ wrong side

$\alpha_i \in [0, C_k] \rightarrow$ on the boundary



(V)

#error

weighted error

$C_1 = C_2 = 1$

3

3

$C_1 = 1, C_2 = 10$

7

16

$C_1 = 10, C_2 = 1$

4

4